

National Yang Ming Chiao Tung University
Institute of Electrical and Control Engineering

Robotics Project: Part 1

Due: 11/21/24 Fall 2024

For a Mitsubishi RV-M2 type six-joint robot manipulator with the following kinematic table

Joint	d(in)	a(in)	α	θ
1	0	120mm	-90°	0°
2	0	250mm	0°	0°
3	0	260mm	0°	0°
4	0	0	-90°	0°
5	0	0	90°	0°
6	0	0	0°	0°

$$-150^\circ \leq \theta_1 \leq 150^\circ, -30^\circ \leq \theta_2 \leq 100^\circ$$

$$-120^\circ \leq \theta_3 \leq 0^\circ, -110^\circ \leq \theta_4 \leq 110^\circ$$

$$-180^\circ \leq \theta_5 \leq 180^\circ, -180^\circ \leq \theta_6 \leq 180^\circ$$

please write a program for the following two transformations:

- input: Cartesian point (n, o, a, p), output: the corresponding joint variables.
- input: joint variables, output: Cartesian point (n, o, a, p) and (x, y, z, ϕ , θ , ψ).